

Technical Report: The Driving Simulation Telemetry Dashboard & Research Suite

I. Introduction: The Intersection of Simulation and Data Science

The **Driving Simulation Telemetry Dashboard** is a sophisticated, high-fidelity software ecosystem designed to serve as a critical bridge between complex vehicle physics engines and the end-user's analytical requirements. In the modern era of sim-racing and automotive engineering, the ability to interpret data is as important as the ability to drive. This application is engineered to interface seamlessly with industry-leading simulators such as BeamNG.drive and Assetto Corsa, utilizing high-speed UDP and shared memory protocols to capture telemetry at frequencies up to 60Hz. Beyond its role as a simple visualization tool, the application functions as a comprehensive research suite, providing a laboratory-grade environment for performance optimization, vehicle health monitoring, and human-factor studies. By transforming raw binary streams into intuitive visual analytics and portable datasets, the dashboard empowers drivers and engineers to uncover the hidden dynamics that define the boundary between control and catastrophe.

II. Real-Time Telemetry and Visual Analytics

A. The Live Dashboard Interface: A Pilot's Perspective

The heart of the application is the real-time dashboard, a high-density information environment designed to minimize cognitive load while maximizing situational awareness. At the center of this interface is the **Friction Circle (G-G Diagram)**, a vital tool that plots lateral versus longitudinal acceleration. This visualization is critical for mastering the "limit" of a vehicle; it allows a driver to see whether they are properly "trail braking"—smoothly transitioning from longitudinal braking force to lateral cornering force—without exceeding the tire's grip capacity. Complementing this is the **High-Fidelity Input Visualizer**, which provides millisecond-accurate feedback on the driver's pedal and steering inputs. This is not merely a display of current state, but a diagnostic tool for identifying technical flaws. For example, it can reveal "pedal overlap," where a driver is inadvertently applying both throttle and brake, a common error that generates excessive heat and reduces cornering efficiency. By visualizing these inputs in real-time, the application helps drivers build the muscle memory required for professional-grade consistency.

B. Vehicle Systems and Tire Thermal Dynamics

Modern racing is a game of management, and this application provides the tools necessary to monitor a vehicle's vital signs under extreme stress. The **Vehicle Health Monitor** tracks the cumulative damage and wear on five critical systems: the engine, transmission, suspension, brakes, and aerodynamics. Each system is modeled on a 0–100% health scale, where degradation is calculated based on factors like high-RPM operation, mechanical impacts, and thermal stress. Simultaneously, the **Tire Dynamics Suite** monitors the temperature and wear of each individual tire. Tire management is often the deciding factor in endurance racing; by providing real-time thermal data, the dashboard allows drivers to adjust their lines or pace to prevent "glazing" the brakes or "overcooking" the tires. This level of detail ensures that every mechanical stress is accounted for, allowing teams to develop strategies based on hard data rather than intuition.

III. The Behavioral Simulation Engine: A Research Powerhouse

A. Procedural Driver Modeling and Synthetic Data

One of the most innovative and technologically advanced features of the suite is its internal **Procedural Simulation Engine**. This module allows the application to function entirely independently of an external game, generating realistic telemetry data through an internal physics and driver model. This is not a simple random number generator; it is a sophisticated procedural system that follows a simulated track layout, modulating throttle, brake, and steering based on a target racing line. This engine is a cornerstone for research, as it allows for the generation of massive datasets under perfectly controlled conditions. It is pa

rticularly valuable for training machine learning models intended for autonomous driving or driver assistance systems, providing a safe and cost-effective alternative to generating thousands of hours of real-world or in-game telemetry.

B. Impairment, Aggression, and Human-Factor Simulation

The simulation engine is further enhanced by its **Behavioral Profile System**, which can emulate a wide spectrum of human driving states. Users can select from profiles such as **Professional**, **Reckless**, **Slow**, **Drunk**, or **High**. Each profile modifies the underlying control logic in specific ways: the "Drunk" profile introduces significant control latency and high-frequency steering "noise" to simulate motor-skill impairment, while the "Reckless" profile prioritizes aggressive throttle application and late braking, often at the cost of vehicle health. These simulations are used to study the "signatures" of impaired driving, providing researchers with the data necessary to develop detection algorithms that can identify erratic behavior before an accident occurs. By quantifying the relationship between a driver's state and the resulting vehicle dynamics—such as increased "jerk" or frequent "oversteer corrections"—the application provides a unique window into the mechanics of human error.

IV. Human-Centric Performance: The Reaction Time Suite

Understanding the driver's cognitive state is as important as understanding the car's mechanical state. To this end, the application includes a dedicated **Reaction Time Testing Suite**. This module utilizes the user's actual sim-racing hardware (the brake pedal) to measure physical response times to visual stimuli. During the test, the application monitors for a "false start" and then records the precise millisecond delay between a visual cue and the initial brake application. This tool is invaluable for assessing driver fatigue during long-duration sessions. By comparing reaction times at the beginning and end of a four-hour stint, drivers can objectively measure their cognitive decline, providing a data-driven justification for pit stops or driver changes. It also allows for the benchmarking of different drivers within a team, identifying who has the sharpest reflexes under pressure.

V. Data Architecture and the Analysis Workflow

The application's data management system is designed for both speed and long-term storage. Utilizing a **SQLite database backend**, every frame of telemetry is logged with high-resolution timestamps. This architectural choice ensures data integrity and allows for complex querying during post-session review. The **Analysis Tab** provides a powerful interface for scrubbing through historical data, where users can overlay different laps or sessions to identify where time was lost. Summary statistics like "Coast Time Percentage" and "Fuel Efficiency" are calculated automatically, providing a high-level view of performance. This workflow transforms raw telemetry from a transient stream of numbers into a permanent library of performance, where every session is a lesson that can be studied and mastered.

VI. The CSV Dataset: A Granular Data Dictionary

A. Core Physics and Kinematic Variables

The CSV export flattens the application's high-frequency SQLite database into a portable, time-indexed format optimized for external processing in environments like Excel, Python (Pandas/NumPy), or MATLAB. Each row represents a discrete "frame" of vehicle state, providing a granular look at the following primary variables:

- * **timestamp**: Recorded in Unix epoch milliseconds, this variable is the foundation for all time-series analysis. It allows researchers to synchronize telemetry with external video feeds or heart-rate monitors and is used to calculate the precise duration between events, such as the time elapsed between a "false start" and a corrective brake application.

- * ****`speed` & `rpm`****: These are the primary indicators of longitudinal performance. Beyond simple velocity tracking, analyzing the relationship between RPM and speed across different gears allows engineers to visualize the vehicle's power band and identify the "optimal shift points" where torque delivery is maximized. Discrepancies between theoretical speed (based on RPM/Gear) and actual speed are used to detect wheel spin or clutch slip.
- * ****`gear`****: This categorical variable tracks the transmission state. In manual simulation mode, analyzing shift times (the duration where `clutch` is active and `gear` is transitioning) provides a metric for driver proficiency. In racing, gear selection analysis reveals whether a driver is utilizing engine braking effectively to stabilize the car during corner entry.
- * ****`gForceX`, `gForceY`, `gForceZ`****: These represent the vector components of the vehicle's acceleration. `gForceX` (Lateral) measures cornering intensity and tire lateral grip; `gForceY` (Longitudinal) measures acceleration and braking force; and `gForceZ` (Vertical) captures the vehicle's "heave," revealing the impact of track bumps, curbs, and aerodynamic downforce at high speeds. Summing these vectors provides a total "G-load" which is used to assess the physical stress on both the driver and the vehicle's structural components.
- * ****`x, y, z`****: These 3D world coordinates allow for the exact reconstruction of the vehicle's trajectory. By plotting these points, researchers can create "heat maps" of the racing line, identifying where a driver is missing apexes or taking inefficient paths. The `y` coordinate (elevation) is particularly useful for analyzing the impact of gravity on braking distances and suspension loading on undulating circuits.

B. Derived Behavioral and Efficiency Metrics: The "Why" Behind the Data

The true analytical power of the application lies in its derived metrics—variables that are not directly provided by the simulator but are calculated by the dashboard's backend to provide a semantic interpretation of the raw physics:

- * ****`jerk\_x` & `jerk\_y`****: Defined as the time-derivative of acceleration (the rate of change of G-forces), jerk is the ultimate proxy for driving "smoothness." High jerk values indicate abrupt, unsettled movements that shock the tire contact patch and destabilize the vehicle's weight distribution. Professional drivers strive to minimize jerk to maintain a "steady state" of grip, whereas impaired or aggressive drivers typically exhibit high jerk values, making this a key variable for safety and behavioral profiling.
- * ****`slip\_angle\_estimate`****: This variable quantifies the angular difference between the vehicle's heading and its actual direction of travel. It is a critical metric for understanding the "friction ellipse" of the tires. By monitoring slip angles, engineers can identify the exact threshold where the car transitions from a "grip" state to a "slip" state, providing a quantitative measure of understeer (front-end wash) or oversteer (rear-end slide).
- * ****`pedal\_overlap`****: This metric calculates the simultaneous application of throttle and brake pedals. While professional "left-foot braking" involves a controlled overlap to stabilize the car, excessive or unintentional overlap is a primary indicator of driver panic or lack of technical discipline. In research, high pedal overlap values during high-speed sections are often correlated with cognitive load or motor-skill degradation.
- * ****`coasting\_time\_pct` & `is\_coasting`****: These variables identify "dead time" where the driver is providing no significant input to either the throttle or brake. In endurance racing, this is a vital efficiency metric; by maximizing "lift and coast" at the end of straights, drivers can significantly reduce fuel consumption with minimal impact on lap times. Conversely, in a qualifying scenario, excessive coasting indicates a lack of aggression and lost performance.
- * ****`oversteer\_correction` & `understeer\_plough`****: These are complex behavioral flags. `oversteer\_correction` detects when a driver applies counter-steering in response to a rear-end slide, measuring both the magnitude of the slide and the speed of the driver's reaction. `understeer\_plough` identifies instances where high steering angles result in no corresponding change in lateral G-force, signifying that the front tires have reached their limit and are "ploughing" straight ahead.

* **`throttle\_delta` & `brake\_delta`**: These measure the "rate of application" for the pedals. A high `throttle\_delta` indicates an aggressive "stomp" on the accelerator, which can induce wheel spin, while a high `brake\_delta` can trigger ABS or lock the wheels. These variables are used to differentiate between the controlled, progressive inputs of a professional and the erratic, binary inputs of an inexperienced or impaired driver.

* **`damage\_engine/transmission/suspension/brakes/aero`**: These variables track the cumulative mechanical cost of a session. By correlating damage rates with specific behavioral profiles (e.g., "Reckless"), researchers can quantify the long-term financial and safety costs of aggressive driving styles.

VII. Conclusion

The **Driving Simulation Telemetry Dashboard** is more than a visualization tool; it is a comprehensive diagnostic and research platform. By merging real-time visual analytics with a sophisticated behavioral simulation engine and a granular data-logging architecture, it provides a holistic view of the driving experience. Whether it is being used to shave tenths of a second off a qualifying lap, manage vehicle health during a 24-hour race, or conduct serious research into the dynamics of impaired driving, the application delivers the precision and depth required for professional-grade telemetry analysis. It stands as a testament to the power of data to inform, improve, and ultimately transform our understanding of the complex relationship between the driver and the machine.

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